ESP32 - ANALOG BOARD

05/11/2026

Variant: [No Variations]

REV. B

Material: JLC04161H-3313

RELEASED OCT-26

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DESIGN CONSIDERATIONS

DESIGN NOTE:
Example text for informational design notes.

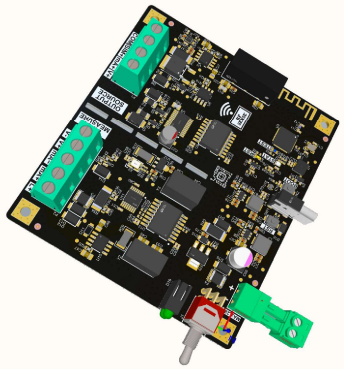
DESIGN NOTE:
Example text for debug notes.

DESIGN NOTE:
Example text for critical design notes.

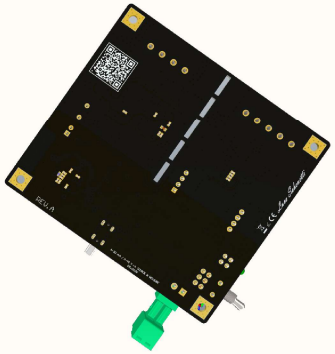
DESIGN NOTE:
Example text for cautionary design notes.

LAYOUT NOTE:
Example text for critical layout guidelines.

TOP VIEW



BOTTOM VIEW



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ESP32 ANALOG BOARD

Portable tester / calibrator for PLC analog I/O - Rev. B - 22/10/2026

PURPOSE

IO TESTER is a portable instrument for commissioning and checking PLC analog I/O channels (e.g. Siemens SM 331 / 6ES7135 family) directly in the field.
It generates the signal a PLC analog input expects, measures the signal a PLC analog output produces, and simulates a 2-wire 4-20 mA transmitter.
It is operated from a phone or PC through a Wi-Fi web interface served by the on-board ESP32-S3.

MAIN FEATURES

- Analog output (source): 0-10 V and 4-20 mA, 12-bit (U5 AD5412).
- Analog input (measure): 4-20 mA and 0-10 V, 24-bit delta-sigma (U11 ADS1220).
- 2-wire transmitter simulation: 4-20 mA current sink powered by the PLC loop.
- L+ 24 V loop supply, current limited to |7 mA, to power external 2-wire transmitters.
- Two galvanically isolated field domains (output and input), isolated from each other and from the MCU / USB side.
- Power from a 5 V USB-C power bank or a 24 VDC field supply, selected with J1.
- Wi-Fi web UI and OTA firmware update (ESP32-S3, 8 MB flash).
- Design target accuracy: +/-0.3 % of full scale.
- Protected field terminals: PPTC / resettable fuses, bidirectional TVS, clamp diodes.

POWER SUPPLY

- P3/P4 - 24 VDC field input (3.81 mm pluggable): F4 mSMD050-60V 0.5 A PTC, D16 SMAJ30CA, reverse-polarity P-MOS Q3 BSP315P (D15 BZT52C15, R47 100K, C70 47n soft-start), second clamp stage R45 4R7 + D17 SMAJ30CA.
- U14 AP63301WU-7 buck 24 V -> 5 V (+5V_FIELD), 500 kHz.
- USB1 USB-C - power bank input (+5V_USB via FB1 120R), CC pull-downs 5K1, ESD protection U15 TPD2E2U06 + D18 SMF6.0CA. Recommended power bank: 5 V, 2 A min.
- J1 DPDT - selects +5V_SYS from FIELD or USB; the second pole tells the MCU the active source (MCU_PWR_FIELD / MCU_PWR_BANK).
- U16 AP63301WU-7 buck 5 V -> 3.3 V (+3V3) for the MCU side.
- 5V_BANK_LVL: 150K / 68K divider to GPIO1 for power-bank voltage monitoring.
- D19 bicolour LED: +5V_FIELD / +5V_USB present.

ISOLATION

- Three ground domains: system GND (MCU, USB, power), DAC iso, ADC iso.
- Digital isolators U9 and U13 NSI8262W1-DSWR: 4 forward / 2 reverse channels, outputs default high.
- Isolated DC/DC: Y1 B0524S-2WR3 (DAC, 24 V), Y2 B0524S-1WR3 (L+ loop, 24 V), Y3 B0505S-1WR3 (ADC, 5 V). Minimum-load bleeders R13 10K (DNP), R29 5K1, R37 470R.
- PCB: 8 mm clearance across each isolation barrier.

FIRMWARE (PLANNED)

- Two-point calibration per range stored in NVS, with calibration date.
- Drift compensation with the ADS1220 internal temperature sensor.
- Self-test by loopback from P1 to P2.
- OTA update over Wi-Fi.

OUTPUT CHANNEL - DAC (ISOLATED, P1)

- U5 AD5412AREZ 12-bit DAC with integrated voltage / current output, SPI through U9.
- Separate terminals: V_OUT 0-10 V and I_OUT 4-20 mA.
- AVDD = +24 V iso from Y1 B0524S-2WR3; AVSS = -4 V from U8 LM27761 inverting charge pump, so the 0-10 V range is specified down to code 0.
- ADC_5V_ISO rail from U7 LM317L (240R / 715R) supplies U8 and U6.
- External R_SET 15K (R23); internal reference REFOUT -> REFIN; DVCC from the internal LDO (also supplies the isolator output side).
- +V_SENSE taken at the TVS node (Kelvin) to compensate the protection drop.
- FAULT read back through U9: open-loop and over-temperature detection.
- V_OUT protection: R14 33R, D5 BAT54S clamp to the rails, D7 SMBJ30CA, F1 JK-nSMD010/60V PPTC.
- I_OUT protection: R17 33R, D9 BAT54S, R18 47R, D10 SMAJ24CA, F2 JK-nSMD010/60V PPTC.

2-WIRE TRANSMITTER SIMULATION (P1 SINK+)

- The PLC analog input powers the loop; the board sinks the programmed current from SINK+ to GND.
- U6 OP184 drives Q2 IRLR3110Z with R16 25.5 ohm sense resistor (Kelvin).
- Setpoint = V_SENSE / 16.4 (R19 15K4 / R20 1K), I_sink = V_SENSE / 418 ohm: 10 V = 23.9 mA.
- 2_WIRE_T_EN (GPIO38, through U9): high = sink OFF (Q1 AO3400A pulls the Q2 gate low). The isolator default-high output keeps the sink off during reset and boot.
- Protection: D6 1N4148W series diode (reverse polarity), D8 SMAJ33CA.

INPUT CHANNEL - ADC (ISOLATED, P2)

- U11 ADS1220 24-bit delta-sigma ADC, SPI through U13 (DOUT/DRDY and DRDY).
- Current input mA+ / mA-: R34 25.5 ohm shunt, RC filter 2 x 4K12 + 27 nF differential + 2n7 common mode, PGA gain 2 (20 mA = 0.51 V). Protection F3 BSMD1812-010-60V 0.1 A, D13 SMAJ33CA.
- Voltage input V+ / V-: divider R40 90K9 / R42 10K (1:10.09), R43 9K09 source balance on AIN3, filter 10 nF differential + 1 nF common mode, gain 1. Protection D14 SMAJ33CA.
- X2 4.096 MHz external oscillator: simultaneous 50 / 60 Hz rejection at 20 SPS.
- Reference: internal 2.048 V (REFP0 also wired to AVDD for ratiometric use).
- Supply: Y3 B0505S-1WR3 -> U12 HT7333 3.3 V (AVDD); DVDD and oscillator through 10R filters.
- L+ loop supply (P2 pin 5): Y2 B0524S-1WR3 24 V iso -> U10 LM317L current limiter, R28 47R (|6.6 mA), D11 protection diode, D12 SMAJ33CA.

PCB AND MANUFACTURING

- 4 layers, 130 x 90 mm, JLCPCB stackup JLC04161H-3313.
- L1 signals / components, L2 continuous GND, L3 power, L4 GND + AVSS island.
- SMT assembly JLCPCB, components LCSC (codes in the BOM).
- Kelvin connections on shunts R16 / R34; AD5412 exposed pad = AVSS (-4 V copper island).

MCU AND USER INTERFACE

- U3 ESP32-S3FN8 (8 MB flash), X1 40 MHz crystal.
- ANT1 PCB antenna with pi matching network C11 1p2 / L2 2n7 / C12 1p5 (35+j0 -> 50 ohm).
- Native USB on USB1 (D+ / D-) for programming and debug.
- SW1 RESET (RC on CHIP_PU); U4 74LVC1G14 Schmitt trigger drives the D4 reset LED. SW2 BOOT on GPIO0.
- D3 RGB LED, D2 blue LED (test board), U1 passive buzzer KLJ-4020 driven by U2 SS8050.

ESP32-S3 GPIO MAP

GPIO0	BOOT_MODE	GPIO1	5V_BANK_LVL
GPIO4	DAC_FAULT	GPIO5	DAC_SDO
GPIO6	DAC_SDIN	GPIO7	DAC_SCLK
GPIO8	DAC_LATCH	GPIO10	ADC_DRDY
GPIO11	ADC_DOUT/DRDY	GPIO12	ADC_SCLK
GPIO13	ADC_CS	GPIO14	ADC_DIN
GPIO17	MCU_PWR_FIELD	GPIO18	MCU_PWR_BANK
GPIO19	USB_D-	GPIO20	USB_D+
GPIO33/34/35	LED R/G/B	GPIO36	LED blue
GPIO37	BUZZER	GPIO38	2_WIRE_T_EN
GPIO46	strap (5K1 to GND)		

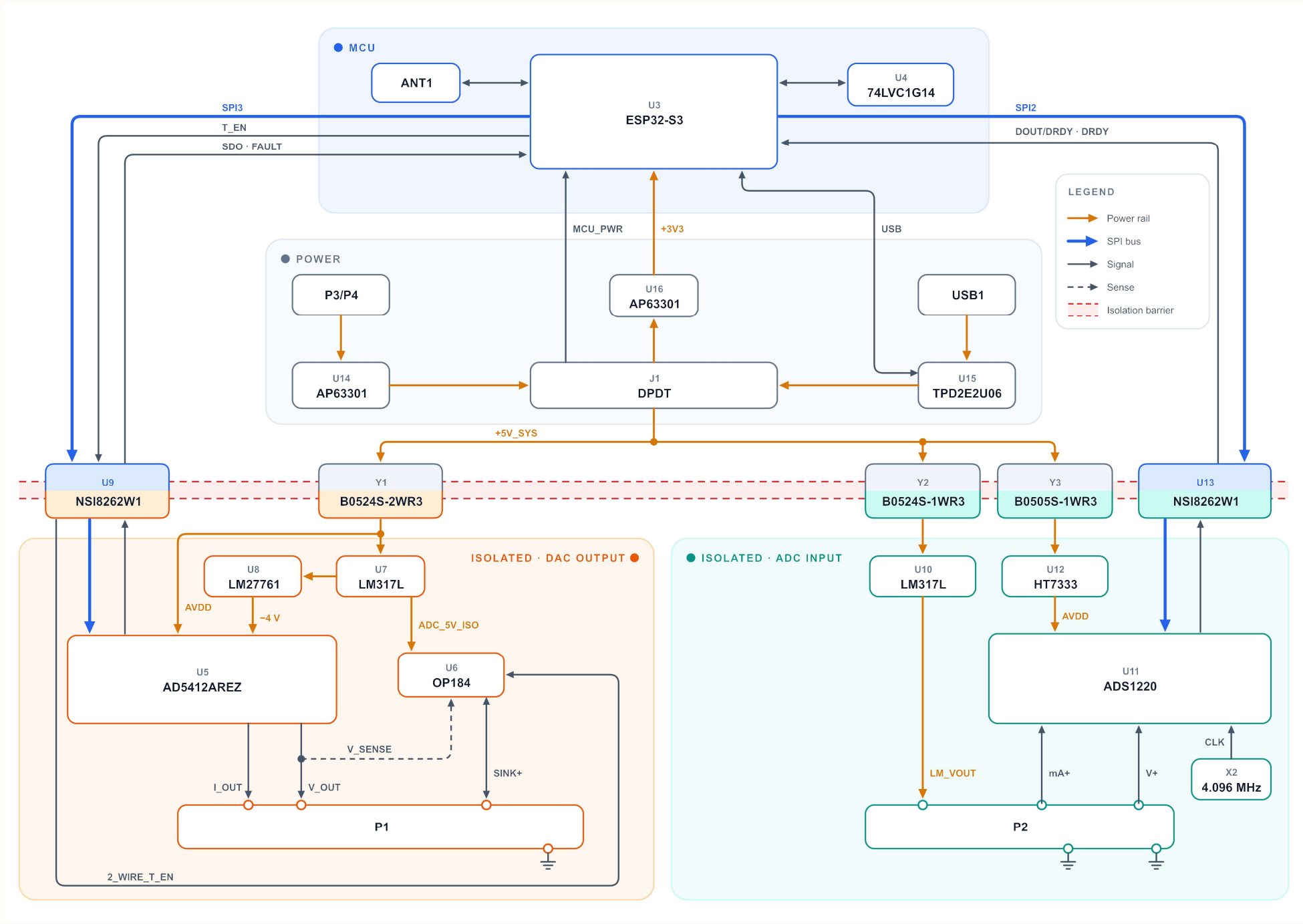
CONNECTOR PINOUT

P1 OUTPUT (WJ500V-5.08-4P): 1 GND (DAC iso) - 2 SINK+ - 3 I_OUT - 4 V_OUT
P2 INPUT (WJ500V-5.08-5P): 1 V- - 2 V+ - 3 mA- - 4 mA+ - 5 L+ (24 V loop supply)
P3/P4 FIELD SUPPLY (3.81 mm): 1 GND - 2 +24 VDC
USB1: USB-C (power bank + native USB data)

TYPICAL CONNECTIONS

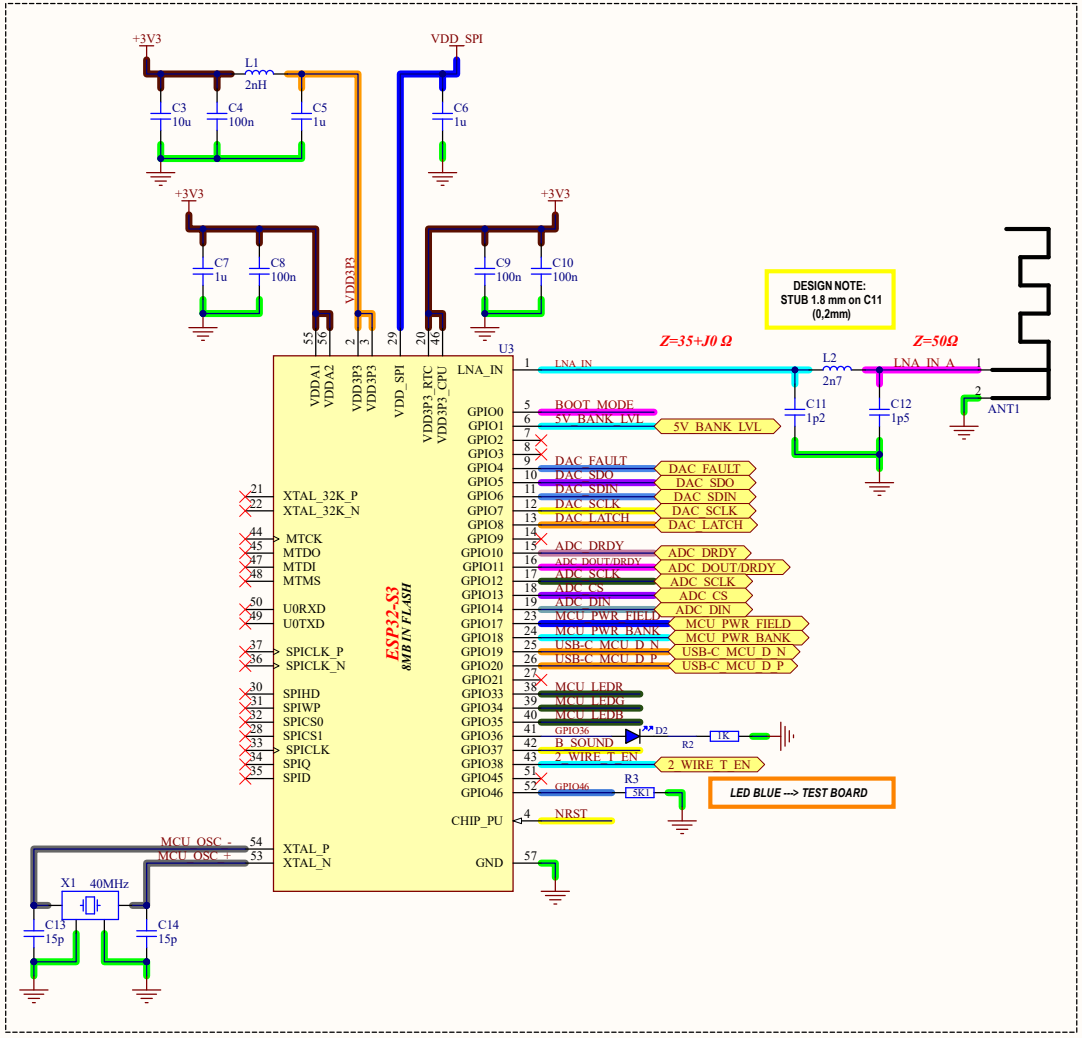
- Test a PLC AI 4-20 mA (4-wire): P1 I_OUT -> AI+, P1 GND -> AI-.
- Test a PLC AI 0-10 V: P1 V_OUT -> AI+, P1 GND -> AI-.
- Simulate a 2-wire transmitter on a loop-powered PLC AI: loop + -> P1 SINK+, loop - -> P1 GND.
- Measure a PLC AO current: AO+ -> P2 mA+, AO- -> P2 mA-.
- Measure a PLC AO voltage: AO+ -> P2 V+, AO- -> P2 V-.
- Read a 2-wire transmitter: P2 L+ -> TX+, TX- -> P2 mA+.

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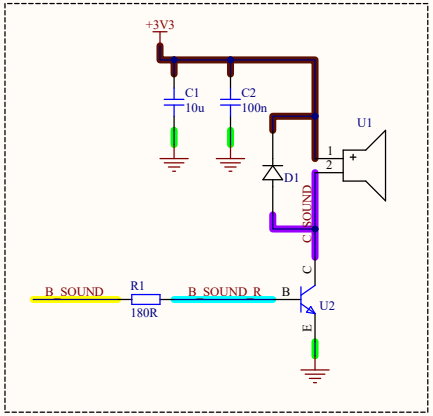


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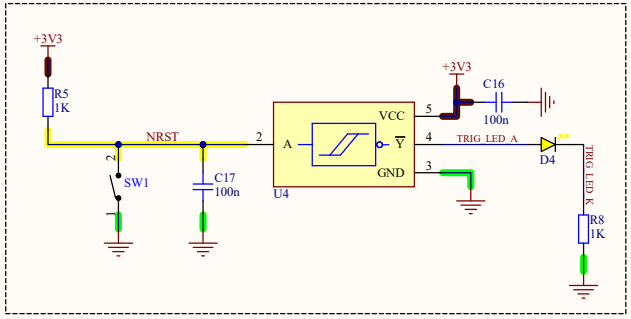
ESP32 MCU



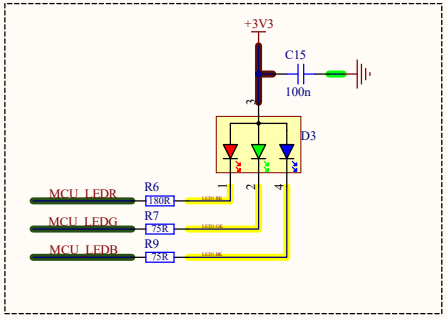
BUZZER



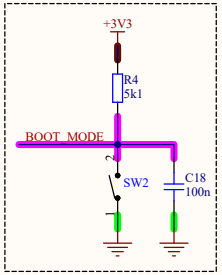
RESET CIRCUIT



LED RGB

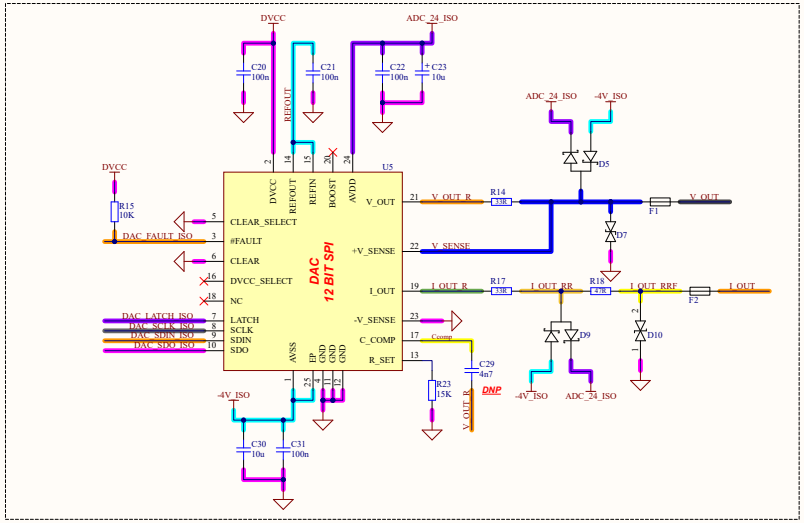


BOOT

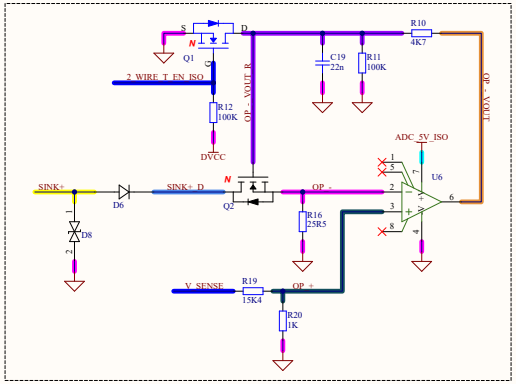


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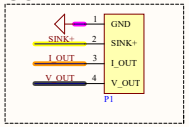
DAC TRASMITTER



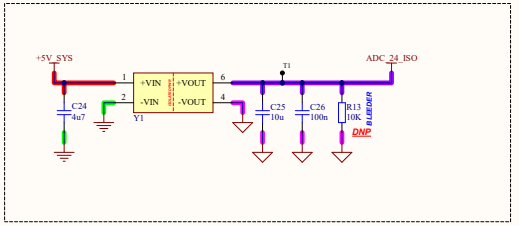
2 WIRE CIRCUIT



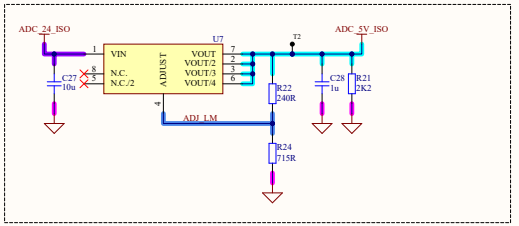
CONN



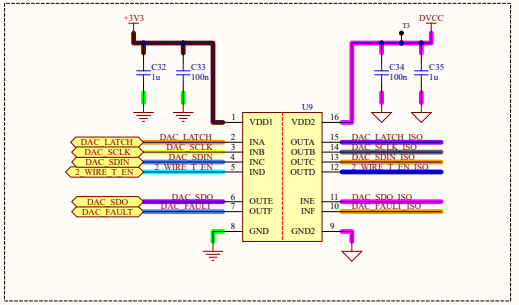
24V PWR ISO



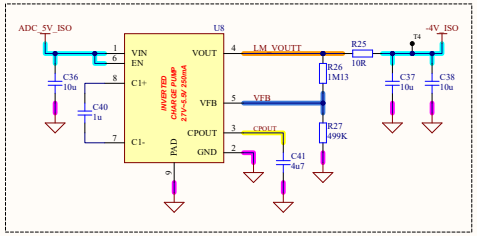
5V PWR ISO



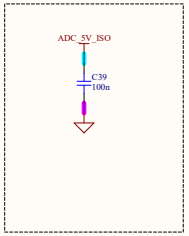
SPI ISO



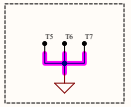
-5V CHARGE PUMP



OPAMP DEC

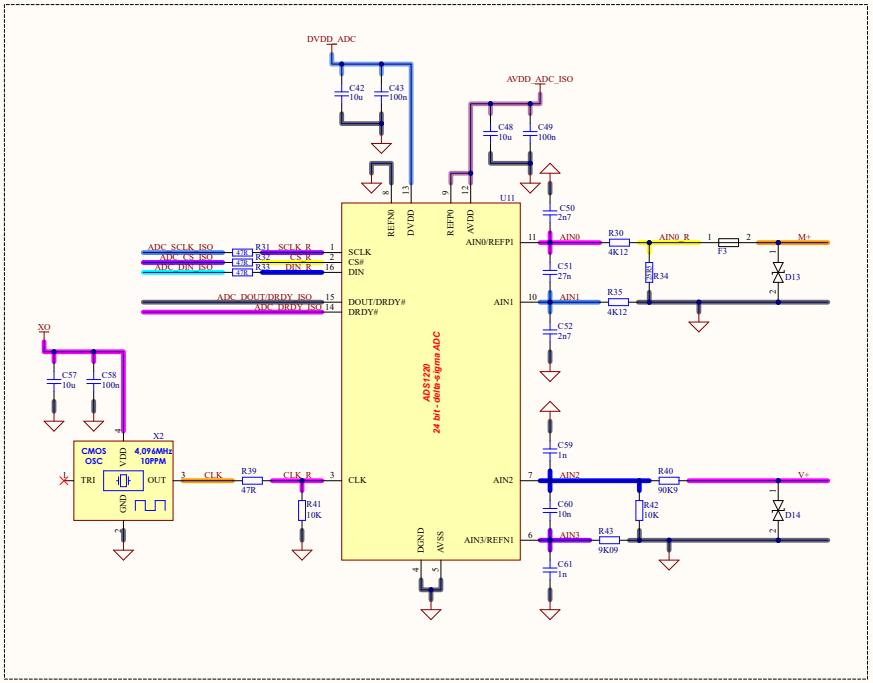


TEST

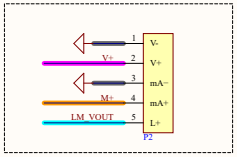


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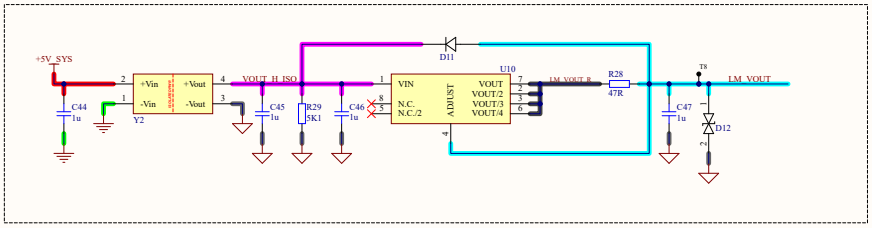
2 - 4 WIRE RECEIVER



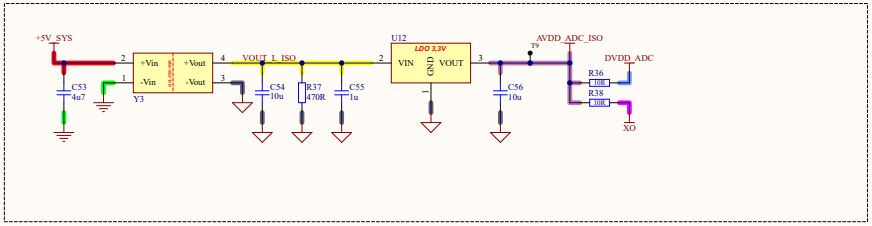
CONNECTOR



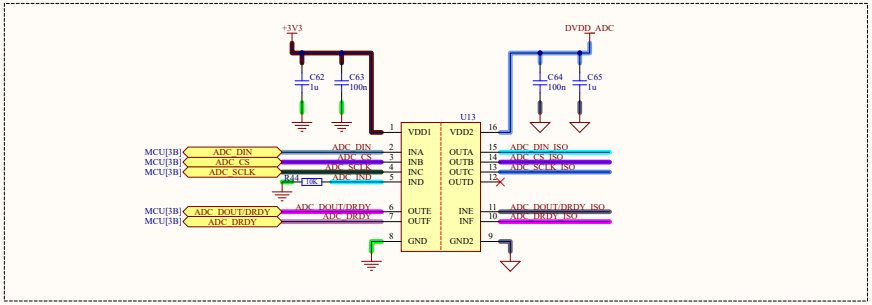
24V PWR ISO



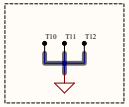
5V PWR ISO



SPI ISO

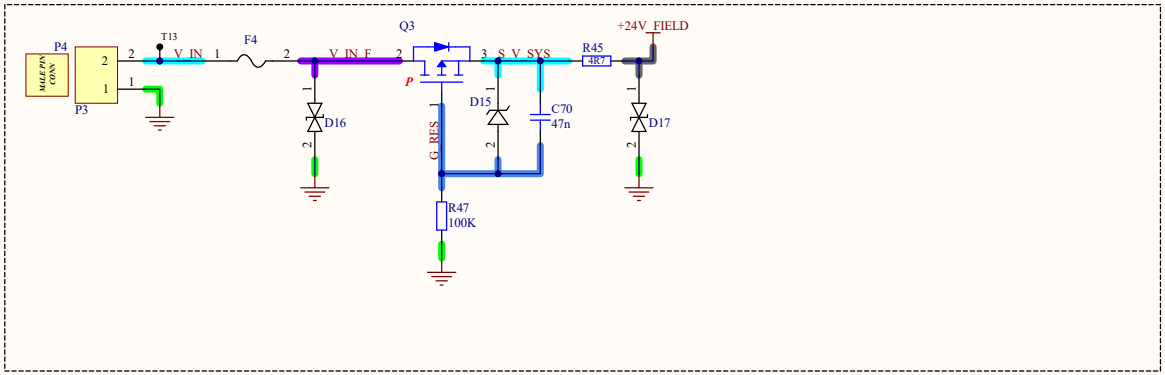


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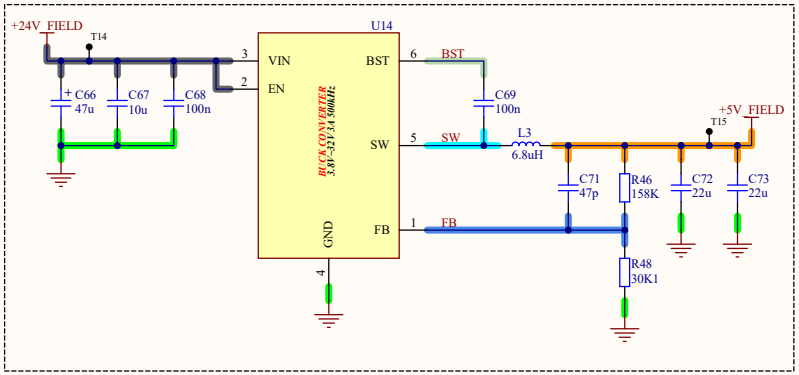


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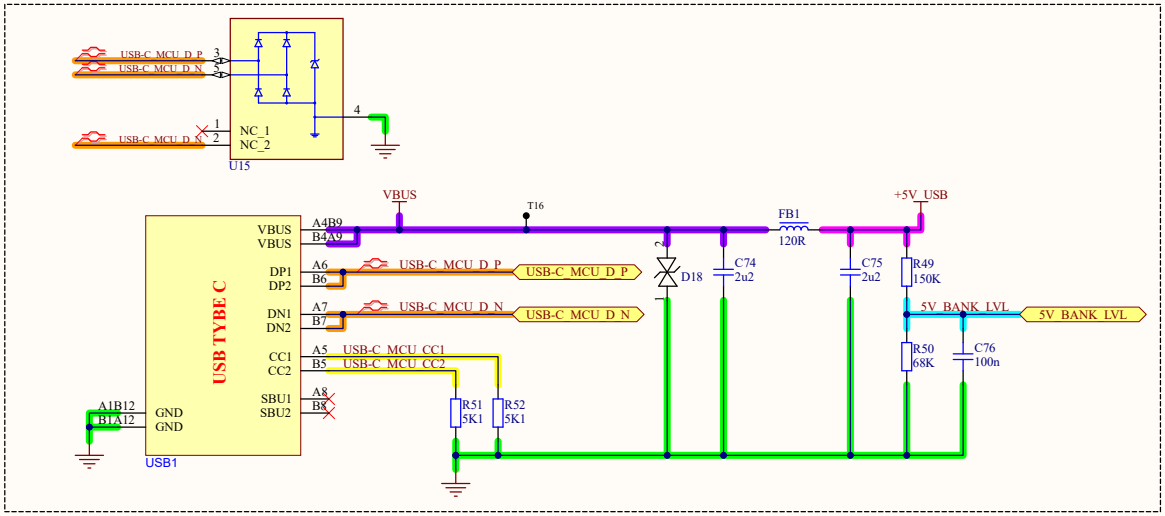
24V FIELD - PROTECTION



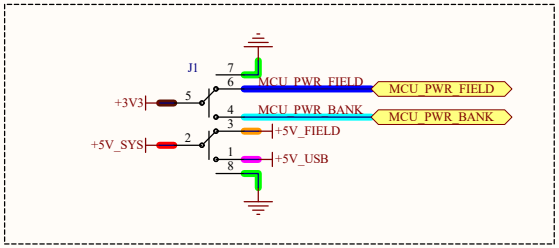
STEP DOWN REG - 5V



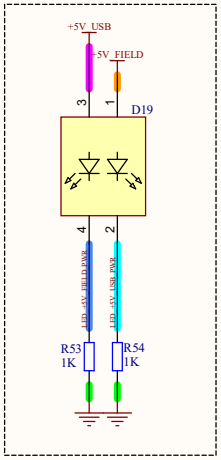
POWER BANK - USB & ESD PROTECTION



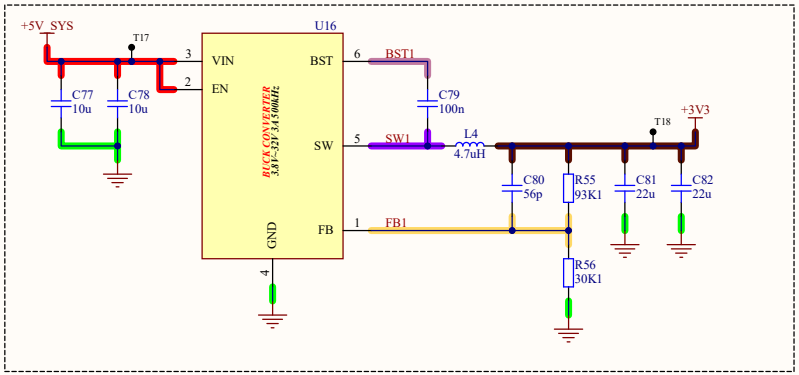
GENERAL - SELECTOR



LED - PWR



STEP DOWN REG



TEST



FIDUCIAL



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